Labeling

In ML we need features (X) and labels (y).

Fixed-time horizon method

Uses triple classification based on fixed threshold (T) at fixed bar interval:

- $y = 1$: price goes up by more than T
- $y = -1$: price goes down by more than T
- $y = 0$: price change stays within $[-T ; T]$

⚠ Method relies on time bars, so poor statistical properties (heavy tails, heteroscedasticity)

Triple-Barrier method

Instead of waiting for a timer to expire, wrap the price path in a "box".

1. Upper horizontal barrier (profit-taking): set as a multiple of dymamic volatility.
   If price hits, $y = 1$ (successful long trade)
2. Lower horizontal barrier (stop-loss) : also set based of volatility.
   If price hits, $y = -1$
3. Vertical barrier (time-expiration) : If price hits, can set $y = 0$, or better is sign of returns on the period.

Learning side AND size

A strategy needs to answer three questions:

1. When to enter?
2. Which direction?
3. How much capital?

⚠ Under these circumstances we cannot know which horizontal barrier is profit-taking, because model doesn't know side yet (long vs short)

Therefore we are REQUIRED to use symmetric barriers.
(solution to problem => META-LABELING)

Meta - Labeling

It's the process of building a secondary model that filters the predictions of a primary model.

- Primary model: idea generator, for example: mean-reversion, MA crossover ... (SIDE)
- Secondary model: risk manager / size controller, looks at market conditions (SIZE)

Advantages:

- Asymmetric barriers restored: say profit-take (3x vol.), stop-loss (1x vol.)
- 3-class to binary problem: secondary model only cares if trade is profitable
  $y \in \{0 ; 1\}$ (pass vs take bet)
- Determines size: models like XGBoost output probability. Can use it to size bet.

⚠ Primary model should focus on recall (True Positive): Catch as many possible trades.
Secondary model should focus on precision: Out of all trades how many were profitable?